

Spark MLlib is regarded as a distributed machine learning framework on top of the Spark Core which, mainly due to the distributed memory-based Spark architecture, is almost nine times as fast as the disk-based implementation used by Apache Mahout.

The various common machine learning and statistical algorithms that have been implemented and included with MLlib are:

- Summary statistics, correlations, hypothesis testing, random data generation
- *Classification and regression*: Supports vector machines, logistic regression, linear regression, naïve Bayes classification
- Collaborative filtering techniques including Alternating Least Squares (ALS)
- Cluster analysis methods including k-means and Latent Dirichlet Allocation (LDA)
- Optimisation algorithms such as stochastic gradient descent and limited-memory BFGS

The latest version is 2.0.1.

Website: <http://spark.apache.org/mllib/>

TensorFlow

TensorFlow is an open source software library for machine learning developed by the Google Brain Team for various sorts of perceptual and language understanding tasks, and to conduct sophisticated research on machine learning and deep neural networks. It is Google Brain's second generation machine learning system and can run on multiple CPUs and GPUs. TensorFlow is deployed in various products of Google like speech recognition, Gmail, Google Photos and even Search.

TensorFlow performs numerical computations using data flow graphs. These elaborate the mathematical computations with a directed graph of nodes and edges. *Nodes* implement mathematical operations and can also represent endpoints to feed in data, push out results or read/write persistent variables. *Edges* describe the input/output relationships between nodes. *Data edges* carry dynamically-sized multi-dimensional data arrays or tensors.

Its features are listed below.

- *Highly flexible*: TensorFlow enables users to write their own higher-level libraries on top of it by using C++ and Python, and express the neural network computation as a data flow graph.
- *Portable*: It can run on varied CPUs or GPUs, and even on mobile computing platforms. It also supports Docker and running via the cloud.
- *Auto-differentiation*: TensorFlow enables the user to define the computational architecture of predictive models combined with objective functions, and can handle complex computations.
- *Diverse language options*: It has an easy Python based interface and enables users to write code, and see visualisations and data flow graphs.

The latest version is 0.10.0.

Website: www.tensorflow.org

Oryx 2

Oryx 2 is a realisation of Lambda architecture built on Apache Spark and Apache Kafka for real-time large scale machine learning. It is designed for building applications and includes packaged, end-to-end applications for collaborative filtering, classification, regression and clustering.

Oryx 2 comprises the following three tiers.

- *General Lambda architecture tier*: Provides batch, speed and serving layers, which are not specific to machine learning.
- *Specialisation on top* which, in turn, provides machine learning abstraction to hyperparameter selection, etc.
- *End-to-end implementation* of the same standard machine learning algorithms as an application (ALS, random decision forests, k-means) on top.

Oryx 2 consists of the following layers of Lambda architecture as well as connecting elements.

- *Batch layer*: Used for computing new results from historical data and previous results.
- *Speed layer*: Produces and publishes incremental model updates from a stream of new data.
- *Serving layer*: Receives models and updates, and implements a synchronous API, exposing query operations on results.
- *Data transport layer*: Moves data between layers and takes input from external sources.

The latest version is 2.2.1.

Website: <http://oryx.io/>

Accord.NET

Accord.NET is a .NET open source machine learning framework for scientific computing, and consists of multiple libraries for diverse applications like statistical data processing, pattern recognition, linear algebra, artificial neural networks, image and signal processing, etc.

The framework is divided into libraries via the installer, compressed archives and NuGet packages, which include Accord.Math, Accord.Statistics, Accord.MachineLearning, Accord.Neuro, Accord.Imaging, Accord.Audio, Accord.Vision, Accord.Controls, Accord.Controls.Imaging, Accord.Controls.Audio, Accord.Controls.Vision, etc.

Its features are:

- Matrix library for an increase in code reusability, and gradual change of existing algorithms over standard .NET structures.
- Consists of more than 40 different statistical distributions like hidden Markov models and mixture models.
- Consists of more than 30 hypothesis tests like ANOVA, two-sample, multiple-sample, etc.
- Consists of more than 38 kernel functions like KVM, KPC and KDA.